

Fisher Information and Delta Method for the Toothpaste Cap Problem

Setup

For the *toothpaste cap problem*, we have

$$\ell(\pi) = \log L(\pi) = k \log \pi + (N - k) \log(1 - \pi)$$

and

$$\hat{\pi} = \frac{1}{N} \sum_{i=1}^N y_i = \text{avg}(y) = \frac{k}{N}.$$

To estimate the variance, we need

$$\widehat{\text{Var}}(\hat{\pi}) \approx [\mathcal{J}_{\text{obs}}(\hat{\pi})]^{-1} = \left[- \left. \frac{d^2 \ell(\pi)}{d\pi^2} \right|_{\pi=\hat{\pi}} \right]^{-1}.$$

Finding $\widehat{\text{SE}}(\hat{\pi})$

Finding derivatives of ℓ

The first derivative is $\frac{d\ell}{d\pi} = \frac{k}{\pi} - \frac{N-k}{1-\pi}$.

The second derivative is $\frac{d^2\ell}{d\pi^2} = -\frac{k}{\pi^2} - \frac{N-k}{(1-\pi)^2}$ (see below).

i Some tedious algebra

$$\frac{d\ell}{d\pi} = \frac{k}{\pi} - \frac{N-k}{1-\pi},$$

$$\frac{d^2\ell}{d\pi^2} = \frac{d}{d\pi} \left(\frac{k}{\pi} \right) - \frac{d}{d\pi} \left(\frac{N-k}{1-\pi} \right) \quad \text{differentiate term by term}$$

$$= k \frac{d}{d\pi} (\pi^{-1}) - (N-k) \frac{d}{d\pi} ((1-\pi)^{-1}) \quad \text{rewrite as powers}$$

$$= k(-1)\pi^{-2} - (N-k)[(-1)(1-\pi)^{-2} \cdot (-1)] \quad \text{power rule + chain rule}$$

$$= -\frac{k}{\pi^2} - (N-k)(1-\pi)^{-2}$$

$$= -\frac{k}{\pi^2} - \frac{N-k}{(1-\pi)^2}.$$

Simplifying $-\ell''(\hat{\pi})$

Evaluate the negative second derivative at $\pi = \hat{\pi}$.

$$-\left. \frac{d^2\ell}{d\pi^2} \right|_{\pi=\hat{\pi}} = \frac{N}{\hat{\pi}(1-\hat{\pi})}$$

Thus,

$$\widehat{\text{Var}}(\hat{\pi}) \approx \left[-\left. \frac{d^2\ell}{d\pi^2} \right|_{\pi=\hat{\pi}} \right]^{-1} = \left[\frac{N}{\hat{\pi}(1-\hat{\pi})} \right]^{-1} = \frac{\hat{\pi}(1-\hat{\pi})}{N}$$

and

$$\widehat{\text{SE}}(\hat{\pi}) \approx \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{N}}$$

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Use the following identities:

- $k = N\hat{\pi}$. $\rightarrow$ Multiply both sides of $\hat{\pi} = k/N$ by N .
- $N - k = N(1 - \hat{\pi})$. $\rightarrow N - k = N - (N \cdot k/N) = N - N\hat{\pi} = N(1 - \hat{\pi})$.
- $\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}$. $\rightarrow$ Common, useful identity.

$$\begin{aligned} -\left. \frac{d^2\ell}{d\pi^2} \right|_{\pi=\hat{\pi}} &= \frac{k}{\hat{\pi}^2} + \frac{N-k}{(1-\hat{\pi})^2} && \text{evaluate at } \pi = \hat{\pi} \\ &= \frac{N\hat{\pi}}{\hat{\pi}^2} + \frac{N(1-\hat{\pi})}{(1-\hat{\pi})^2} && \text{use } k = N\hat{\pi}, N - k = N(1 - \hat{\pi}) \\ &= N\left(\frac{1}{\hat{\pi}} + \frac{1}{1-\hat{\pi}}\right) && \text{cancel a factor in each term} \\ &= N\left(\frac{\hat{\pi} + (1-\hat{\pi})}{\hat{\pi}(1-\hat{\pi})}\right) && \text{use } \frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} \\ &= \frac{N}{\hat{\pi}(1-\hat{\pi})} && \text{simplify} \end{aligned}$$

Delta Method

We often report odds

$$\tau(\pi) = \frac{\pi}{1 - \pi}, \quad \widehat{\text{odds}} = \tau(\hat{\pi}) = \frac{\hat{\pi}}{1 - \hat{\pi}}.$$

Finding the derivative of $\tau(\pi)$

For the delta method, we need the derivative of $\tau\pi$.

$$\tau'(\pi) = \frac{1}{(1 - \pi)^2}.$$

i Some tedious algebra

$$\begin{aligned} \tau(\pi) &= \frac{\pi}{1 - \pi} \\ \tau'(\pi) &= \frac{1 \cdot (1 - \pi) - \pi \cdot (-1)}{(1 - \pi)^2} && \text{quotient rule} \\ &= \frac{1 - \pi + \pi}{(1 - \pi)^2} = \frac{1}{(1 - \pi)^2}. && \text{combine terms} \end{aligned}$$

Plugging into the delta method equation

Recall that $\widehat{\text{Var}}(\tau(\hat{\theta})) \approx (\tau'(\hat{\theta}))^2 \widehat{\text{Var}}(\hat{\theta})$.

Applying this, we obtain

$$\widehat{\text{Var}}(\widehat{\text{odds}}) \approx \frac{\hat{\pi}}{N(1-\hat{\pi})^3}.$$

Taking the square root,

$$\widehat{\text{SE}}(\widehat{\text{odds}}) = \sqrt{\frac{\hat{\pi}}{N(1-\hat{\pi})^3}}.$$

i Some tedious algebra

$$\begin{aligned} \widehat{\text{Var}}(\widehat{\text{odds}}) &\approx (\tau'(\hat{\pi}))^2 \widehat{\text{Var}}(\hat{\pi}) && \text{delta method} \\ &= \left(\frac{1}{(1-\hat{\pi})^2} \right)^2 \cdot \frac{\hat{\pi}(1-\hat{\pi})}{N} && \text{use } \tau'(\pi) = \frac{1}{(1-\pi)^2}, \quad \widehat{\text{Var}}(\hat{\pi}) = \frac{\hat{\pi}(1-\hat{\pi})}{N} \\ &= \frac{1}{(1-\hat{\pi})^4} \cdot \frac{\hat{\pi}(1-\hat{\pi})}{N} && \text{square the derivative} \\ &= \frac{\hat{\pi}}{N(1-\hat{\pi})^3} && \text{cancel one factor of } (1-\hat{\pi}) \text{ and rearrange} \end{aligned}$$